

BUILDING YOUR OWN PRIVATE CHATGPT

USING:

DOCKER, OLLAMA,
OPEN WEBUI, Z-IMAGE TURBO



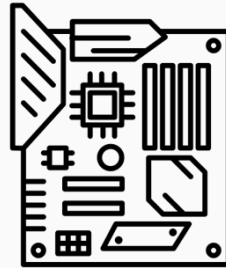
CHEAP(ISH) USABLE

AI @ Home

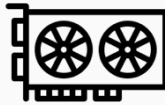
USING OLD OUTDATED

JUNKS

(FOR MY FAMILY USE)



8th Gen i5
16 GB RAM
RTX 3060 12GB



CPU 2017
GPU 2021



MAUNGS.CO

M

Lwin Maung

Managing Director

Accolade Solutions



x 13

Azure Foundry AI ♦ Windows Development



MIN
LWIN

MAUNGS.CO

M

Min Maung

Managing Director

M3 Technical Services



x 10

Azure Foundry AI ♦ Windows Development



MIN MAUNGS.CO

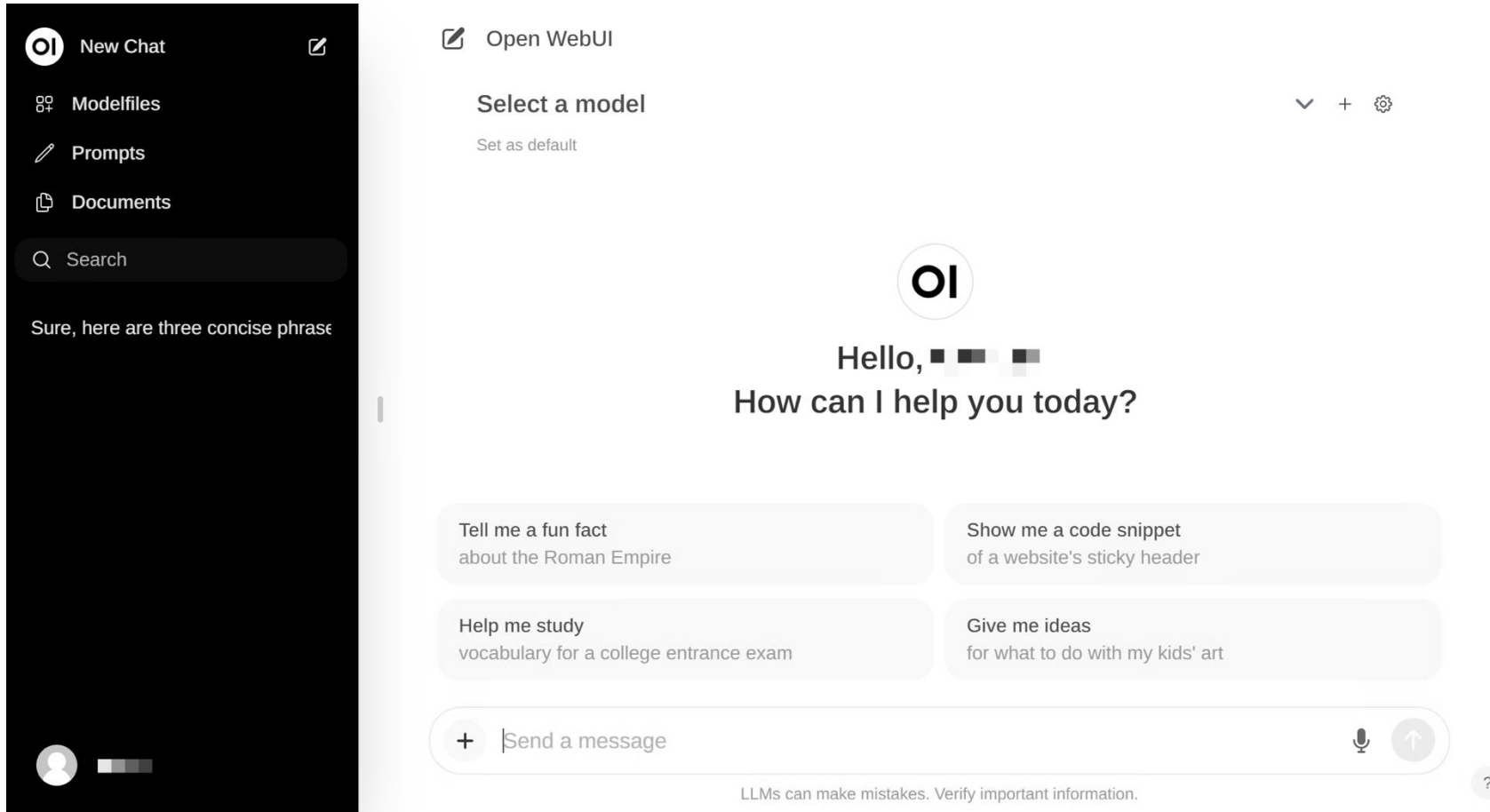
M

Items needed for today

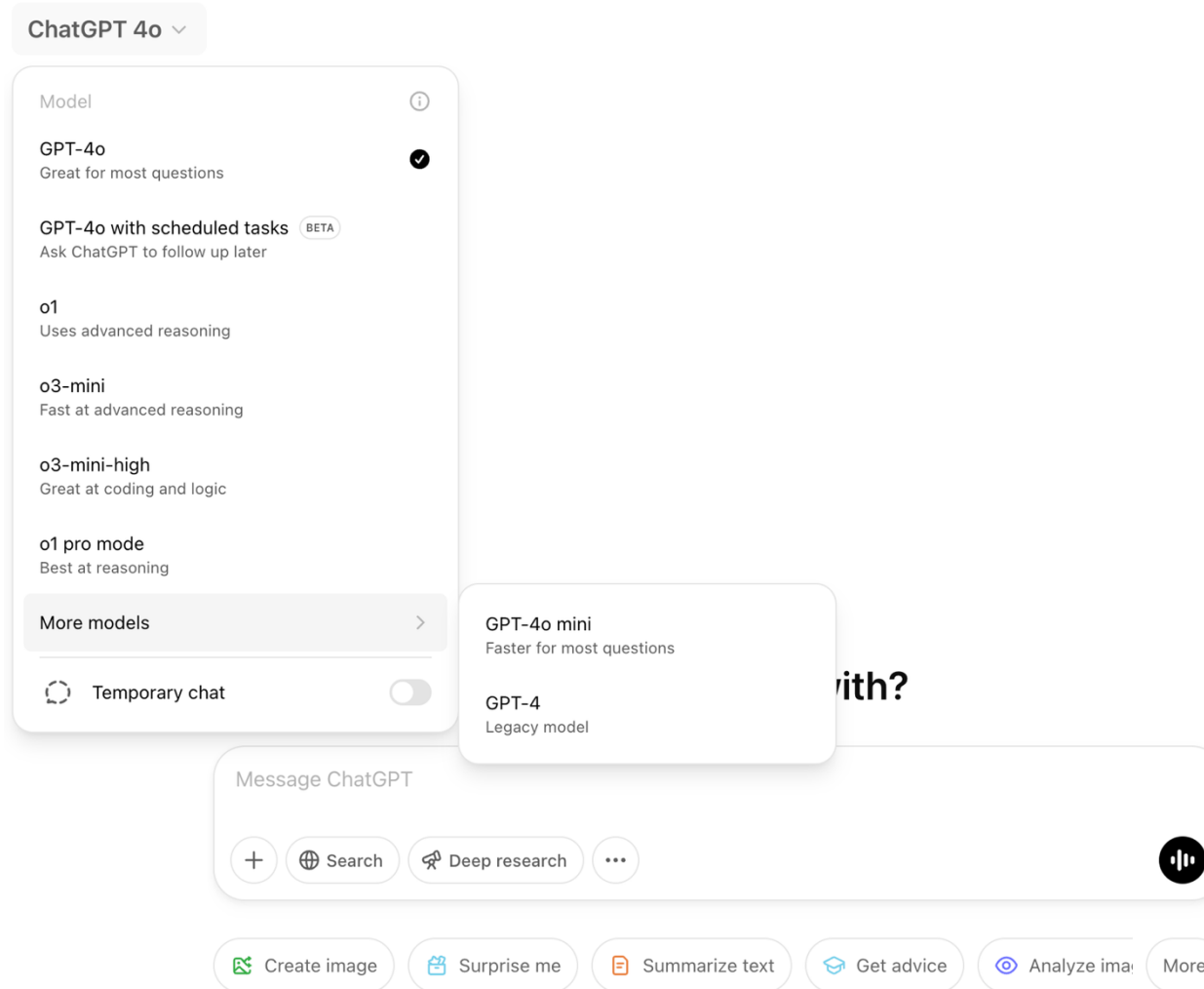
- Ollama
- Phi4-Mini
- Docker
- Working Computer

+ Working Computer with >12GB of VRAM with enough CUDA cores

We will build this and run it...



Are we NVTs? Why? ChatGPT is only \$20.00/month



Data Matters

You get to go home with...



Let's all have a dream...



Reality...



Reality...





What this sessions will not cover



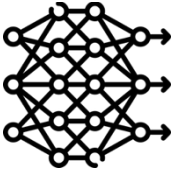
In depth model
comparison



OS
comparison



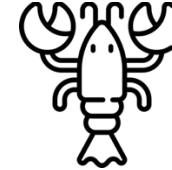
Best
practices



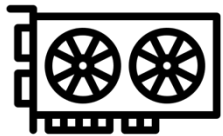
Training /
Weights / RAGs



Infrastructure
production use



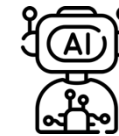
Open
Claw



In depth GPU
comparison



IT
security



AI
Assistants

How does (AI) computer work?

 Explain Quantum Computing

pp

Prompt Processing

Tokenization

text is broken into tokens (small chunks of words)

Prompt embedding

tokens are converted into numerical vectors

Context processing

the model analyzes the entire prompt

tg: tokens/sec

Token Generation

How AI produces responses

After the prompt is processed, the AI begins generating tokens.

A **token** is a piece of text.

The Generation Loop

For every token generated:

the model predicts probabilities
chooses the next token
appends it to the response
repeats

This loop runs **many times per second**.

Context Window

How much text the AI remembers

The **context** window is the amount of text the model can process at once.

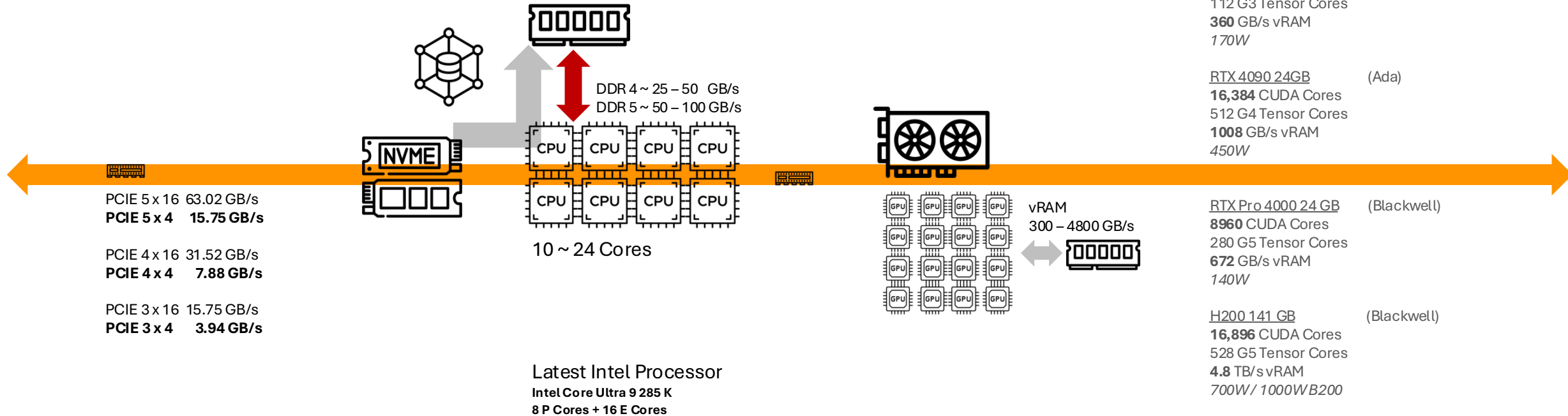
Typical modern LLM memory context
32K–128K tokens

How does (AI) computer work?



5-10+ tokens/s

30-100+ tokens/s





This may not be what you are looking for...



Daily use equipment



96 – 128 GB



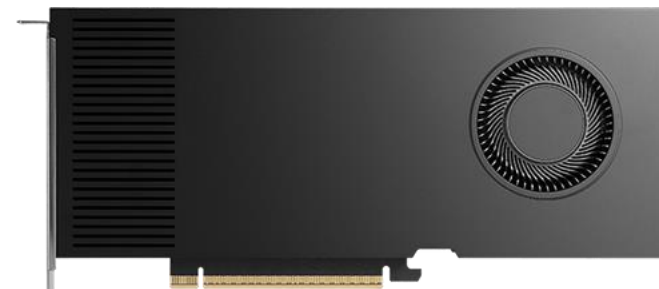
96 GB



128 GB



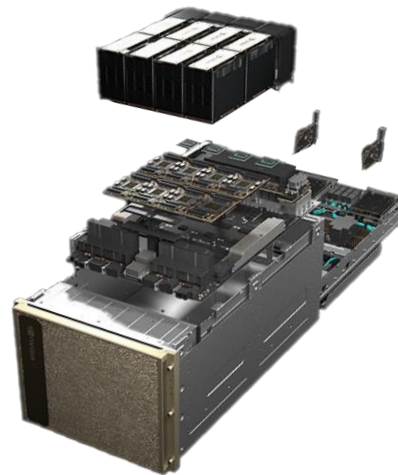
96 GB



24+ GB / GPU

Cost Analysis

DGX Spark	RTX 4090	Nvidia A/H100	Clusters	Cloud Services	Cloud Services
128GB	24GB	40-80GB	8 x Nvidia A100 / H100	Ada Azure ML	H200
\$3,999	\$3,000	\$25,000 - \$45,000	\$19,835/month	\$22.3486/hour	\$95.824/hour
240 W	600 W	700 W	700 W x 8	?	?



Hardware Comparison

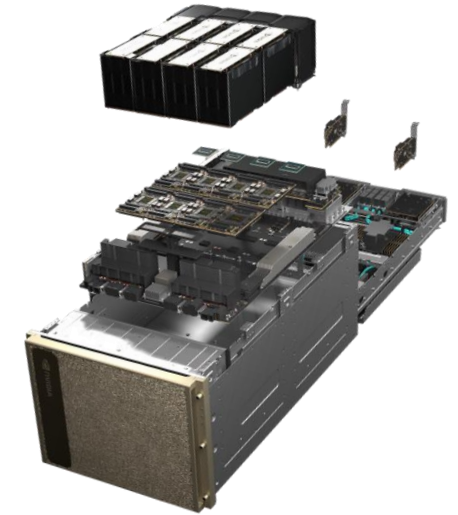
	DGX Spark	H200 	H100 	RTX 5090 	RTX 4090 
GPU Architecture	Nvidia Blackwell	Nvidia Blackwell	Nvidia Hopper	Blackwell**	Ada Lovelace
GPU Memory	128GB unified system memory	140GB	80GB	32GB	24GB
AI Performance	1 PFLOP (FP4)	4 PFLOP (FP8)	898 TFLOP (FP8)	104 TFLOP (FP32)	83 TFLOP (FP32)
System Power	240 W	600 W	700 W	600 W	600 W
Price	\$3,999	\$45,000+	\$25,000+	\$3,500	\$3,000



* Prices may vary and fluctuate depending on vendor

Let's recap

I CAN BUY THESE



Distance & Speed
(TG)



Conversations
(PP)



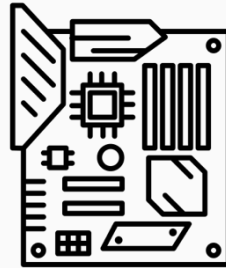
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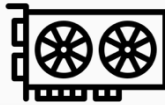
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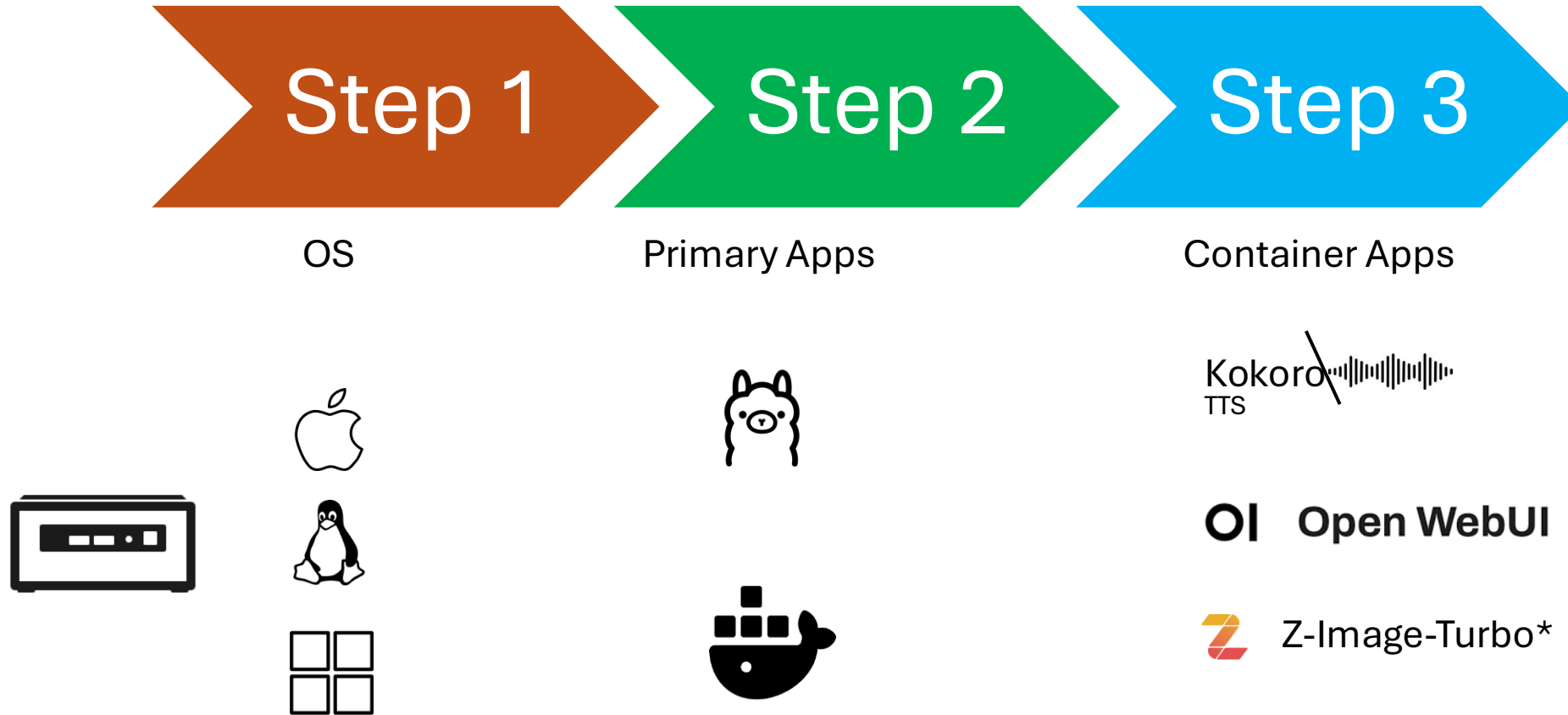
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CPU 2017
GPU 2021



What's the plan guys?



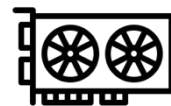
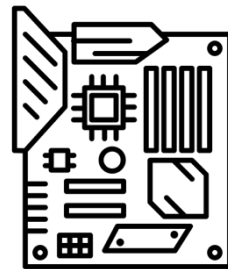
POC > MVP > DAILY DRIVER > PERSONAL SERVER

NUC 10i5
16 GB RAM
iGPU only



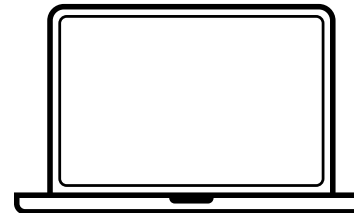
CPU 2019

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16 GB RAM
RTX 3060 12GB



CPU 2017
GPU 2021

MacBook Pro
M3 Max 16 Cores (12/4)
128GB Unified RAM



MinisForum Ai 9
AMD



DGX Spark
NVIDIA



M3 Ultra
Apple

Let's Build

<https://tinyurl.com/cvcc26-maungs>

<https://tinyurl.com/zimagestarter>



Contact Us



Lwin Maung

Microsoft MVP | Board Advisor | Digital Innovator |
Speaker | Author



Min Maung

Tech Visionary | IoT, Mobile, AI Expert |
Transformation Catalyst | Microsoft MVP | IoT, Clou...

